Name: A.Mohammed Kalifa

Reg.no:192411078

Course Name: Computer Networks

Course Code: CSA0714

1. Internet Connectivity for a Remote Rural Branch Office — 10 Marks

Recommended Solution

For a remote rural location where wired broadband is unavailable, the best solution is **4G/5G wireless broadband as the primary connection with satellite internet as a backup**, depending on local availability.

1. Internet Connection

- Use **4G/5G Fixed Wireless Access (FWA)** if cellular coverage is available.
- A minimum of **50–100 Mbps** is recommended for cloud applications and video conferencing.
- Use a **satellite connection as backup** to maintain business continuity.
- Configure automatic **failover** from 4G/5G to satellite if the primary connection fails.

2. Networking Devices

- **5G/4G modem/router** – connects the office to the Internet.
- **Business-grade firewall/router** – manages traffic, VPN, NAT and security.
- **Managed Gigabit switch** – connects computers, printers, servers and access points.
- **Wireless Access Points (APs)** – provide Wi-Fi connectivity.
- **VPN gateway** – provides a secure connection to headquarters.
- **UPS** – protects networking devices from power interruptions.

3. Security Mechanisms

- Establish an **IPsec/SSL VPN** between the branch and headquarters.
- Use a **next-generation firewall (NGFW)** with intrusion prevention.
- Enable **WPA3-Enterprise** for wireless access.
- Implement **multi-factor authentication (MFA)** for important services.
- Use antivirus/endpoint protection and regular software updates.
- Apply access-control rules so employees can access only required resources.

4. Bandwidth Management

Since wireless/satellite bandwidth can be limited:

- Use **Quality of Service (QoS)** to prioritize video conferencing and business applications.
- Block or limit non-business traffic such as streaming.
- Use cloud applications efficiently through caching and compression where possible.
- Monitor bandwidth usage regularly.

5. Reliability

- Use dual Internet connections with **automatic failover**.
- Provide UPS backup for routers, switches and access points.
- Select business-grade networking equipment.
- Monitor the network continuously.

6. Cost and Scalability

- 4G/5G is generally cheaper and easier to deploy than installing wired infrastructure.
- Satellite provides connectivity when terrestrial networks are unavailable.
- The design can later be upgraded to **5G, fiber or higher-capacity satellite**.
- Additional switches and APs can be added as the office expands.

Conclusion

A **4G/5G primary connection + satellite backup**, protected by a firewall and VPN, provides a good balance of **bandwidth, reliability, security and cost**. QoS and automatic failover ensure uninterrupted business operations, while the design remains scalable for future growth.

2. Wireless Network Design for a University with 3,000+ Students — 10 Marks

Recommended Solution

The university should use a **centrally managed dual-band Wi-Fi network** with carefully positioned access points, preferably using **Wi-Fi 6/6E** technology.

1. Requirement Analysis

The network must support:

- More than **3,000 simultaneous users**.
- Classrooms, laboratories, libraries and hostels.
- High-density environments.
- Secure authentication.
- Minimum interference.
- Limited AP deployment because of budget constraints.

2. Access Point Placement

APs should not simply be installed at equal distances.

- Perform a **wireless site survey** before installation.

- Place APs in high-density locations such as classrooms, libraries and laboratories.
- Use ceiling-mounted APs for better coverage.
- Hostels should have APs distributed according to room and floor density.
- Avoid placing APs near concrete walls, electrical equipment and other sources of interference.
- Use directional antennas where appropriate.

3. Frequency Band Selection

5 GHz:

- Use as the primary band for students.
- Provides higher speed and more non-overlapping channels.
- Has less interference than 2.4 GHz.

2.4 GHz:

- Use mainly for older devices and IoT equipment.
- Provides greater range but has more interference.

6 GHz:

- If Wi-Fi 6E-compatible devices are available, use it for high-performance users.
- It provides additional spectrum and reduces congestion.

4. Channel Planning

- Use automatic or centralized channel management.
- Avoid assigning the same channel to nearby APs.
- Use appropriate channel widths, such as **20 MHz in high-density areas**, to reduce interference.
- Adjust transmission power so neighboring APs do not excessively overlap.

5. Authentication and Security

Use **WPA3-Enterprise** with **802.1X/RADIUS authentication**.

Create separate networks/VLANs for:

- Students
- Faculty/staff
- Guests
- University IoT devices

Guest users should have Internet-only access and should not access internal university systems.

6. Centralized Management

Use a **wireless LAN controller/cloud management system** to:

- Monitor APs.
- Manage channels and power.
- Detect interference.
- Balance users between APs.
- Detect unauthorized access points.
- Apply security policies centrally.

7. Performance Improvement

Use **band steering** to move capable devices toward 5 GHz/6 GHz.

Use **load balancing** so that one AP does not become overloaded while nearby APs remain unused.

8. Cost Optimization

Since the number of APs is limited:

- Prioritize high-density areas.
- Use high-capacity Wi-Fi 6/6E APs rather than installing many low-capacity APs.
- Use centralized management to optimize existing APs.
- Expand AP deployment later based on usage statistics.

Conclusion

A **Wi-Fi 6/6E dual/tri-band network with optimized AP placement, 5 GHz/6 GHz priority, centralized channel management, VLANs and WPA3-Enterprise authentication** provides a balance between **coverage, performance, security and cost** for more than 3,000 students.

3. Secure Network Architecture for a Financial Institution — 10 Marks

Recommended Solution

The financial institution should implement a **segmented, defense-in-depth network architecture** using VLANs, firewalls, strong authentication and intrusion detection/prevention.

1. Network Segmentation

Divide the network into separate security zones:

- **Employee VLAN** – normal employee systems.
- **Server VLAN** – internal application and database servers.
- **Customer/Online Banking DMZ** – publicly accessible banking services.
- **Management VLAN** – network and security administration.
- **Guest VLAN** – Internet-only access.
- **Database VLAN** – highly protected customer databases.

This prevents an attacker who compromises one network segment from easily accessing sensitive systems.

2. Firewall Deployment

Use **next-generation firewalls (NGFWs)**.

A firewall should be placed:

- Between the Internet and DMZ.
- Between the DMZ and internal network.
- Between sensitive internal VLANs where required.

Configure:

- Stateful inspection.
- Application filtering.
- Intrusion prevention.
- URL filtering.
- Logging and monitoring.

Only necessary traffic should be permitted.

3. DMZ for Online Banking

Public-facing banking servers should be placed in a **DMZ**.

Example:

Internet → Firewall → DMZ → Internal Firewall → Application Servers → Database

This prevents direct Internet access to the internal database.

4. Authentication

Use strong authentication mechanisms:

- **Multi-factor authentication (MFA)** for employees and administrators.
- **802.1X authentication** for network access.
- Strong password policies.
- Role-Based Access Control (RBAC).
- Privileged Access Management for administrators.

Employees should receive only the permissions necessary for their jobs.

5. Intrusion Detection and Prevention

Deploy **IDS/IPS** to detect:

- Malware.
- Port scanning.
- Unauthorized access.
- Suspicious network traffic.
- Exploitation attempts.

IPS can automatically block known malicious traffic.

6. Encryption

- Use **TLS/HTTPS** for online banking.
- Use VPN/IPsec for remote employees and branch connections.
- Encrypt sensitive information stored in databases.
- Use secure protocols instead of unencrypted protocols.

7. Monitoring and Logging

Implement centralized security monitoring using a **SIEM system**.

It can collect:

- Firewall logs.
- Authentication logs.
- IDS/IPS alerts.
- Server logs.
- User activity.

Security teams can detect unusual activity quickly.

8. Performance Considerations

Security should not significantly reduce network performance.

Therefore:

- Use high-performance firewalls.
- Use hardware-accelerated encryption where available.
- Apply security inspection mainly to relevant traffic.
- Use VLANs to reduce unnecessary broadcast traffic.
- Regularly monitor network performance.

9. Compliance and Cost

The design supports financial-sector security requirements through:

- Strong authentication.
- Encryption.
- Least-privilege access.
- Network segmentation.
- Audit logs.
- Continuous monitoring.

To control costs, existing managed switches can be used for VLAN segmentation, while security appliances are prioritized at critical network boundaries.

Conclusion

A **segmented network with VLANs, DMZ, multiple firewall layers, MFA, RBAC, IDS/IPS, encryption and centralized monitoring** provides strong protection for customer information and online banking services. It balances **security, performance, regulatory compliance and cost** while allowing future expansion.